

Calculus for Biology and Medicine, 3rd Edition, Claudia Neuhauser

12.1 # 1, 5, 11, 16, 25, 29, 30, 34, 36, 37, 38, 46

1. $5 \cdot 2 \cdot 4 = 40$
 ↑ ↑ ↑
 ways to choose fertilizer ways to choose light levels ways to choose replicates

5. $3 \cdot 7 \cdot 4 = 84$
 ↑ ↑ ↑
 ways to choose beverage ways to choose cereal ways to choose fruit

11. order matters, so
 $P(7, 7) = \frac{7!}{0!} = 5040$
 ↑ ↑
 number of books space on bookshelf

16. order matters, so
 $P(10, 4) = \frac{10!}{3!} = 5040$
 ↑ ↑
 number of songs time for songs

25. $\frac{9!}{4!5!} = 126$
 ↑ ↑ ↑
 number of ways to arrange 4 A's number of ways to arrange 5 B's number of ways to arrange 9 letters

29. (a) exactly 2 red balls
 $C(5, 2) = 10$
 ↑ ↑
 number of red balls number of balls picked

exactly 2 blue balls
 $C(4, 2) = 6$
 ↑ ↑
 number of blue balls number of balls picked

exactly 1 of each
 $5 \cdot 4 = 20$
 ↑ ↑
 ways to choose red ways to choose blue

(b) $C(9, 2) = \frac{9!}{2!7!} = 36$ $10 + 6 + 20 = 36$
 ↑ ↑
 number of balls number of balls picked

30. $\frac{12!}{5!4!3!} = 27720$
 ↑ ↑ ↑
 number of ways to arrange a group of 5 children number of ways to arrange 4 children number of ways to arrange 3 children

34. number of subsets with 0 elements number of subsets with 1 element number of subsets with 2 elements ...

$$\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \dots + \binom{n}{n} = \sum_{k=0}^n \binom{n}{k} = \left(\sum_{k=0}^n \binom{n}{k} x^k y^{n-k} \right) = (x+y)^n$$

$$= \sum_{k=0}^n \binom{n}{k} (1)^k (1)^{n-k} = (1+1)^n = 2^n$$

36. $4 \cdot (4-2)! = 8$
 ↑ ↑
 number of seats for Paula number of ways to arrange Gloria and Jenny

37. $\binom{5}{3} + 2 \cdot \binom{5}{2} = 30$
 ↑ ↑ ↑
 number of ways to choose the 5 people can serve with anyone number of people who do not want to serve with each other number of ways to choose 5 people who can serve with anyone in the remaining 2 spots

38. (a) $\binom{9}{3} \cdot \binom{6}{3} = 1680$
 ↑ ↑
 number of ways to choose the first committee number of ways to choose the second committee

46. $C(4, 2) \cdot C(4, 3) = 12$
 ↑ ↑
 number of ways to draw two aces number of ways to draw three kings

(b) $\binom{9}{3} \cdot \binom{9}{3} = 7056$
 ↑ ↑
 number of ways to choose the first committee number of ways to choose the second committee